Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2025-26

Class:	**TE**	**Semester:**	**VI**
Course Code:	**CSL601**	**Course Name:**	**Machine Learning Lab**

Name of Student:	**Shravani Sandeep Raut**
Roll No. :	**40**
Experiment No.:	**07**
Title of the Experiment:	**To implement McCulloch Pitts Model for ANN.**
Date of Performance:	**26/02/2026**
Date of Submission:	**12/03/2026**

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty : Mrs. Sneha Yadav

Signature :

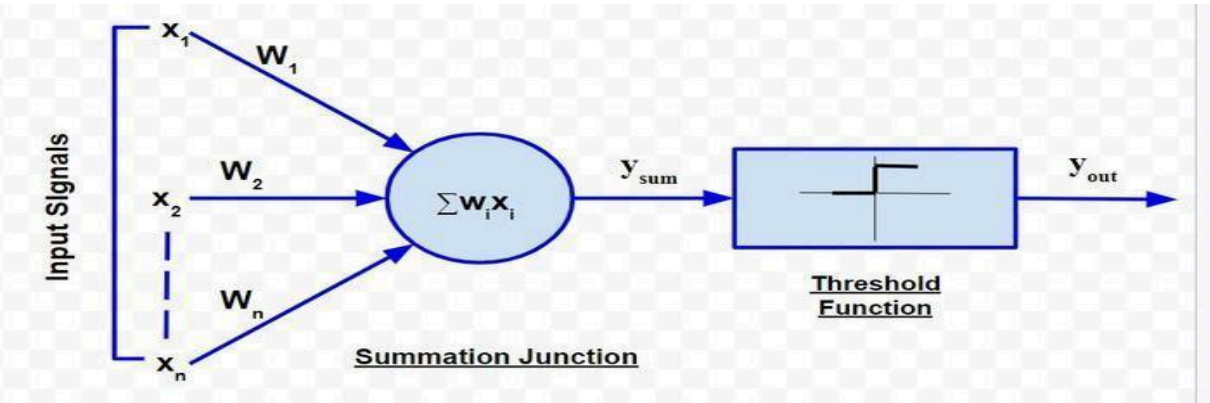
Date:

Aim: To implement McCulloch Pitts Model for ANN.

Objective: Able to design a neural network and use activation function as learning rule defined by McCulloch Piits Model.

Theory:
McCulloch-Pitts Model of Neuron

The McCulloch-Pitts neural model, which was the earliest ANN model, has only two types of inputs — **Excitatory and Inhibitory.** The excitatory inputs have weights of positive magnitude and the inhibitory weights have weights of negative magnitude. The inputs of the McCulloch-Pitts neuron could be either 0 or 1. It has a threshold function as an activation function. So, the output signal y_{out} is 1 if the input y_{sum} is greater than or equal to a given threshold value, else 0. The diagrammatic representation of the model is as follows:



Simple McCulloch-Pitts neurons can be used to design logical operations. For that purpose, the connection weights need to be correctly decided along with the threshold function (rather than the threshold value of the activation function).

Example:

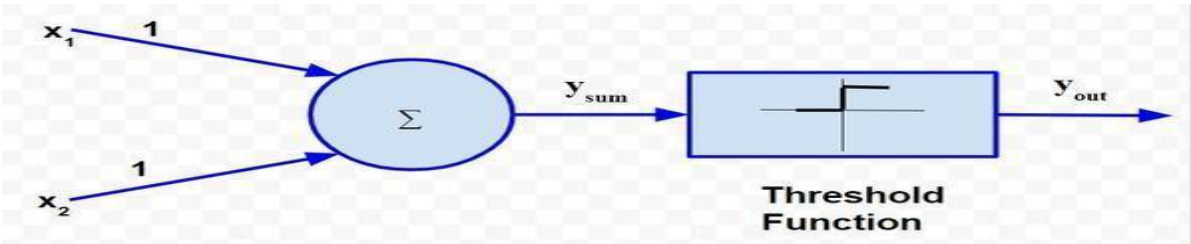
John carries an umbrella if it is sunny or if it is raining. There are four given situations. I need to decide when John will carry the umbrella. The situations are as follows:

- First scenario: It is not raining, nor it is sunny
- Second scenario: It is not raining, but it is sunny
- Third scenario: It is raining, and it is not sunny
- Fourth scenario: It is raining as well as it is sunny

To analyse the situations using the McCulloch-Pitts neural model, I can consider the input signals as follows:

- X_1: Is it raining?
- X_2 : Is it sunny?

So, the value of both scenarios can be either 0 or 1. We can use the value of both weights X_1 and X_2 as 1 and a threshold function as 1. So, the neural network model will look like:



Truth Table for this case will be:

Situation	x1	x2	ysum	yout
1	0	0	0	0
2	0	1	1	1
3	1	0	1	1
4	1	1	2	1

So,

$$y_{sum} = \sum_{i=1}^{2} w_i x_i$$

$$y_{out} = f(y_{sum}) = \begin{cases} 1, x \geq 1 \\ 0, x < 1 \end{cases}$$

The truth table built with respect to the problem is depicted above. From the truth table, I can conclude that in the situations where the value of y_{out} is 1, John needs to carry an umbrella. Hence, he will need to carry an umbrella in scenarios 2, 3 and 4.

Implementation:

```
# General MCP neuron
def mcp_neuron(inputs, weights, theta):
    net = sum(x*w for x, w in zip(inputs, weights))
    return 1 if net >= theta else 0

#Test Data
inputs_list = [(0,0), (0,1), (1,0), (1,1)]

print("AND Gate")
for inputs in inputs_list:
    output = mcp_neuron(inputs, weights=[1,1], theta=2)
    print(inputs, "->", output)

print("\nOR Gate")
for inputs in inputs_list:
    output = mcp_neuron(inputs, weights=[1,1], theta=1)
    print(inputs, "->", output)
```

Output -

```
AND Gate
(0, 0) -> 0
(0, 1) -> 0
(1, 0) -> 0
(1, 1) -> 1

OR Gate
(0, 0) -> 0
(0, 1) -> 1
(1, 0) -> 1
(1, 1) -> 1
```

Conclusion:

1) What is the use of McCulloch Pitts model?
   The **McCulloch–Pitts model** is used to simulate a simple artificial neuron that performs binary logical operations like AND and OR.It forms the foundation of neural networks and helps in understanding how modern artificial intelligence models work.

2) How it will be used to develop ANN?
   The **McCulloch–Pitts model** is used to develop ANN by acting as the basic building block (artificial neuron).By connecting many MCP neurons in layers, we can form an Artificial Neural Network that performs complex decision-making and pattern recognition tasks.